

Co-Supervised Master Theses

by Alex Jung, Assoc. Prof. for Machine Learning

June 14, 2026

Aalto (129 total)

Ongoing (11)

#	Author	Title	Industry	Recording
1	G. Argyrou	Federated Learning for Vision Tasks in Edge-AI Systems		
2	J. Haapanen	Improving the quality of maritime data using artificial intelligence and machine learning methods		
3	A. von Hertzen	Interpretability-Driven Regularization for Deep Weather Forecasting		video
4	M. Holmström	Diagnostics, debugging and analysis tools for ML enabled 5G telecommunication systems		
5	M. Malkki	Vehicle value modelling		
6	S. Mättö	Forecasting seasonality in financial data with machine learning methods		
7	A. Nygård	Agent-System Interface Designs for Enterprise Applications: A Comparative Empirical Study		
8	A. Pääkkilä	Learning Active Constraint Sets with Machine Learning for Constrained Optimization		
9	I. Viita	Automated Validation of Construction Documentation: A Machine Learning Approach Using Industry Data		video
10	H. Välimäki	Generating synthetic 3D food objects for the development of automatic dietary assessment		
11	I. Zalesskii	Bid Optimization in Real-Time Bidding Display Advertising		

Completed (118)

#	Author	Title	Date	Industry	Recording
1	B. Zheng	Exploring the structure in deep networks: Group, manifold and category theory	Dec. 2025		
2	V. Niemi	Diagnosing and improving a maintenance contract churn prediction model	Dec. 2025		video
3	A. Hämäläinen	Improving retrieval-augmented generation with LLM-as-a-judge evaluation	Nov. 2025		
4	S. Monira	FedSock & FedBlue: Scalable decentralized federated learning systems with generalized total variation minimization	Nov. 2025		
5	A. Pavlyuk	Federated learning with multi-layer perceptron for sensor drift compensation in direct-ethanol fuel cells	July 2025		video
6	S. Rachidi	Explainable machine learning in cancer survival prediction — A case study of myeloma patients	July 2025		
7	C. Fait	Designing a data pipeline for total column greenhouse gas measurements with portable EM27/SUN spectrometers	July 2025		
8	L. Tuominen	Robust tap gesture recognition on wearable devices with domain adaptation	July 2025		
9	O. Lauronen	Unsupervised semantic matching of support tickets	May 2025	Polycon Oy	
10	T. Salminen	Contract review with large language models	Apr. 2025		video
11	E. Kohvakka	The role of analyst sentiment in excess return forecasting, and its implications for investors	Apr. 2025		
12	A. Sarkima	Graph-based anomaly detection: Evaluating Logs2Graphs with non-benchmark logs	Mar. 2025		
13	J. Naveed	Optimized code generation in BIM with retrieval-augmented LLMs	Feb. 2025		
14	P. Wiwatphon-thana	Machine Learning Assisted Dynamic Scheduling for Energy Efficient Serverless Cloud Workloads	Jan. 2025	Ericsson Oy	
15	Q. Luong	Proof of Concept for FedRelax on Kubernetes: An Implementation Guide	Jan. 2025		
16	P. Dang	Model-Agnostic Personalized Federated Learning using Adaptive Client Selection	Jan. 2025		video
17	L. Loukamo	The effect of language on perceived ability to understand machine learning concepts	Dec. 2024		
18	I. Tarpila	Training Machine Learning Models for E-sports Spatiotemporal Time-Series Forecasting	Nov. 2024		

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19	I. Lahtinen	Binary classification of invoice documents based on value added tax	Nov. 2024	KPMG Oy Ab	
20	R. Kausiala	Forecasting Retail Investor Fund Flow Using Machine Learning	Nov. 2024		video
21	V. Toivonen	Determining User Preference Profiles from Email And User Engagement Data	Oct. 2024	Vibemetrics Oy	video
22	M. Hilvo	The effect of privacy enhancing technologies on the quality of predictive models in healthcare	Oct. 2024	VTT	
23	A. Bhat	Classifying Scam E-Commerce Shops with Supervised Learning	Oct. 2024	F-Secure	
24	A. Rodimov	Security system based on object tracking by video stream	Aug. 2024		video
25	A. Shuianova	Kinship verification between two people by photos	Aug. 2024		
26	J. Rantanen	Large Language Models in Mental Health Support	July 2024		video
27	R. Mård	Backdoor attacks on large transformer-based regression model	July 2024		video
28	A. Ghazal	Zero-shot Machine Unlearning using GANs	July 2024	Nokia	
29	K. Nguyen	Image Similarity Assessment for Product Quality Assurance	July 2024		
30	D. Suman	Deep Learning Methods for Demand Time Series Forecasting	June 2024	Zalando	
31	A. Manninen	Spatiotemporal Traffic Accident Prediction Using Deep Learning Models	May 2024		video
32	S. Hirvonen	Comparison of data-driven models for building energy load forecasting	May 2024	VTT	
33	L. Ban	Neural Motif Counting in Uncertain Graphs	May 2024		
34	R. Virtanen	Explainable AI Techniques in Trustworthy Object Detection	May 2024	TietoEVERY Oyj	video
35	T. Kulokoski	How well can Machine Learning teach Humans about Machine Learning?	Mar. 2024		video
36	S. Facchini	Prediction of fuel usage consumption in the airline industry	Mar. 2024	Finnair	
37	L. Veneranta	Optimization of Web Page Advertisements using Contextual Bandits	Jan. 2024	Sanoma Media	
38	J. Li	Empirical Emissions Modeling using Machine Learning	Jan. 2024	aurobay.com	
39	K. Izadi Garmaseh	Frequency Offset Estimation Using Deep Learning	Jan. 2024	Nokia	
40	S. Chowdhury	Computer vision-assisted safety alert system for industrial settings in private 5G networks	Jan. 2024	TietoEVERY Oyj	
41	T. Kontola	Predicting User Web Behaviour with Machine Learning Methods	Jan. 2024	Columbia Road Oy	

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42	R. Siljander	Extreme gradient boosting methods for covariate forecasting of housing market demand in Finnish postal code areas	Dec. 2023	Alma Talent Oy	
43	J. Jäkärä	From Candles to Ticks - Improving financial backtesting accuracy	Dec. 2023	Aekos Trading Oy	
44	X. Landa Oregi	MLOps data ingestion pipeline for reciprocal benefit between customer and provider	Nov. 2023	Huawei	
45	T. Brumani	Microservices-based autonomous anomaly detection for mobile network observability	Sept. 2023	Ericsson Oy	video
46	G. Rivi	Software Defined Networking Controlled Energy Optimization through Traffic Prediction on Microwave Access Network	Sept. 2023	Ericsson Finland	video
47	Á. García Gutierrez	Anomaly Detection on Osmosis Trades	Sep. 2023	Numia Data	
48	Z. Liu	Deep Learning based method for Fire Detection	Sept. 2023	Detectium	
49	M. Bogdanova	Contextual bandits for staffing in consulting companies: An exploration of personalized decision making	Sept. 2023		
50	Z. Zhou	Building information modeling connection recommendation based on machine learning using multimodal information	Aug. 2023		video
51	Y. Sarcheshmehpour	Application of Reinforcement Learning in Electrical Machine Design	Aug. 2023	ABB Oy	video
52	L. Garcia Tejada	Applying Machine Learning to Forecast Formula 1 Race Outcomes	Aug. 2023		video
53	C. Segercrantz	Experimental evaluation of record linkage algorithms in a secure banking environment	Aug. 2023	Nordea Bank Oyj	
54	H. Wang	Material Capture and Generative Rendering with Phenomenological Reflectance Models	Aug. 2023	Huawei	
55	G. Jiang	Parallel Training of Neural Networks in 6G L1	Aug. 2023	Nokia Oy	
56	K. Lasocki	Deep learning for continuous symbolic melody generation conditioned on lyrics and initial melodies	Aug. 2023		
57	T. Vanhala	Data-driven xVA exposure calculation for a portfolio of interest rate swaps	May 2023	Nordea Markets	
58	A. Agisheva	Ethics in machine learning publications: Peer-review analysis using NLP methods	May 2023		
59	R. Tikkanen	Clustering fitness tracker data to correct interdevice differences in energy expenditure estimation	May 2023	Fjuul	
60	H. Vu	Deep learning-based Mammography Image Segmentation	Mar. 2023	Planmed	

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61	S. Johansson	Classification of Purchase Invoices to Analytic Accounts with Machine Learning	Jan. 2023	SprintIT	video
62	T. Sormunen	Pallet Detection in Warehouse Environment	Jan. 2023	Wartsila	
63	J. Himanen	Towards a data-driven circular economy: predicting material streams in the construction industry	Jan. 2023		
64	B. Fazekas	Distractive driver behaviour detection with spiked neural networks	Nov. 2022		
65	T. Rahman	Intrusion Detection system based on Deep Learning	Aug. 2022		
66	T. Gyabaah	Artificial intelligence to support NFTs creation: Comparison of Machine learning algorithms to detect fraud in artwork	Jul. 2022		
67	J. Lillfors	Networked Federated Learning	Jul. 2022		
68	Á. C. Barcsa-Szabó	Feature-based Approaches for Ethical News Personalization	Jul. 2022	Sanoma Media Finland	
69	C. Molinero Ranera	Multi-label classification of a hydraulic system using Machine Learning	Jul. 2022		video
70	V. Petrutiu	Exploring Transformers and Degradation Methods in the Super Resolution Field	Jul. 2022	Huawei	
71	P. Truong	Crown-of-Thorns Starfish detection by state-of-the-art YOLOv5	Jul. 2022		
72	Y. Huang	Text analysis of novel coronavirus pneumonia based on federal deep learning	June 2022		
73	C. Ozen	A collaborative approach for large-scale Electricity consumption using Federated Learning	June 2022		
74	P. Prinsén	Robust Gas pressure control using Neural Networks	Jan. 2022	Wartsila Finland Oy	
75	E. Hattula	Transfer Learning Technology for Building Extraction from Orthophotos and Open-Source Data	Jan. 2022	National Land Survey of Finland	
76	A. Channabasaiah	Applying machine learning methods to predict taxi pickups using historical taxi data	Jan. 2022		video
77	R. Hellström	Aspect Based Sentiment Analysis in Finnish	Jan. 2022	Crowst Oy	
78	A. Orre	Pedestrian movement analysis from drone perspective	Dec. 2021		
79	P. Vijayakrishnan	Semi-supervised machine learning techniques for infant motility classification	Oct. 2021		

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80	K. Ariko	Increasing the safety in the proximity of the mobile working machines: a study of detecting people	Oct. 2021	Epec Oy	
81	M. Uutaniemi	Extraction of labeled fields from images of structured documents	Aug. 2021		
82	K. Kutlu	Machine Learning based Chaos Engineering for Cloud-Native Microservice Architectures	Aug. 2021	Ericsson	
83	M. Leinonen	Federated Multi-task Learning over Networked Data	June 2021		
84	J. Seppälä	Application of machine learning to link click predictions in Facebook Family of Apps advertising	2021		
85	M. Afteniy	Predicting time series with Transformer	May 2021		
86	Z. Mohammadi	Better Utilization of Relational Data in Machine Learning	May 2021	Lamia Oy	
87	T. Nguyen	Applying Machine Learning to Develop Black-box Control Model of Active Double-Skin Facade	Jan. 2021		
88	P. Pyrrö	AIR: Aerial Inspection RetinaNet for Land Search and Rescue Missions	Jan. 2021	Accenture	
89	T. Kokkonen	Classifying Restaurant Menu Items With Supervised Learning	Jan. 2021		video
90	I. Vikström	Deep reinforcement learning approach for HVAC control	Dec. 2020	TietoEVERY Oy	video
91	D. Tokmurzina	Road marking condition monitoring and classification using deep learning for city of Helsinki	Oct. 2020		
92	C. Dikmen	Application of Contextual Bandits Models in a Supervised Learning Setting	Aug. 2020		
93	J. Laiho	Recognizing Thoughts from Bioelectric Patterns? A Brain-Computer Interface with Deep Learning	Aug. 2020	Accenture Liquid Studio (NL)	video
94	X. Zhang	Diagnostic and Prognostic Analysis Optimization of Field Problems for EV Charging Stations	Aug. 2020	ABB	video
95	M. Mishin	Anomaly Detection Algorithms and Techniques for Network Intrusion Detection Systems	Aug. 2020	Ericsson	video
96	K. Klemets	Forecasting Hourly Parking Occupancy with Multiple Seasonalities	Aug. 2020	City of Helsinki	
97	T. Valentijn	The Practical Applicability of a CNN for Automated Building Damage Assessment	June 2020	Red Cross NL	
98	T. Hämmäinen	Clustering IoT devices for network intrusion detection systems	May 2020	Ericsson	
99	J. Moisala	Optimizing the mark-up of foreign exchange derivative contracts using machine learning	May 2020		

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100	J. Nieminen	Framework for application of machine learning algorithms in telecommunications	Mar. 2020	Nokia Oy	
101	L. Kolehmainen	A web scraping system for extracting news articles	Dec. 2019	Vainu Finland Oy	
102	M. Torres Porta	Anti-Money Laundering system based on customer behavior	Aug. 2019		
103	T. Wiro	Market influence on purchase prices in procurement	June 2019	Sievo	
104	D. Baad	Automatic Job Skill Taxonomy Generation For Recruitment Systems	June 2019	VXT Research Oy	
105	J. Eskonen	Deep Reinforcement Learning in Automated User Interface Testing	May 2019	Ericsson	
106	A. Moskalev	Demand forecasting for fast-moving products in grocery retail	May 2019	Relex	
107	K. Karapetyan	Process Mining of Automation Services with Long Short-Term Memory Neural Networks	March 2019	Posti Group Oyj	
108	H. Ambos	Semi-Supervised Learning over Complex Networks	Mar. 2019		
109	J. Kahles Bastida	Applying Machine Learning to Root Cause Analysis in Agile CI/CD Software Testing Environments	Jan. 2019	Ericsson	
110	A. Shehata	Cellular Network Average User Throughput-Downlink Prediction by Machine Learning	Dec. 2018	Nokia	
111	O. Abramenko	Graph signal sampling via reinforcement learning	Nov. 2018		
112	M.O. Nasir	Supervised Learning in Lighting Control Systems	Oct. 2018		
113	D. Wu	Unsupervised Learning for Lighting Control System	Oct. 2018	Helvar Oy	
114	N. Pokhrel	Drone Obstacle Avoidance and Navigation Using Artificial Intelligence	May 2018	Nokia	
115	D. Koskenniemi	Do financial networks improve the explanatory power of the Fama-French factors? A comparison of propagation algorithms on stock market returns	Mar. 2018		
116	S. Basirian Jahromi	Compressed Sensing for Big Data Over Complex Networks	Jan. 2018		
117	A. Mara	A Comparative Analysis of Graph Signal Recovery Methods for Big Data Networks	Oct. 2017		
118	Y. Gao	Graphical Model Selection in Big Data Application	Dec. 2016		

TU Wien (1 total)

Completed (1)

#	Author	Title	Date	Industry	Recording
1	B. Kausl	Channel aware inference based on the Fisher information	2012		